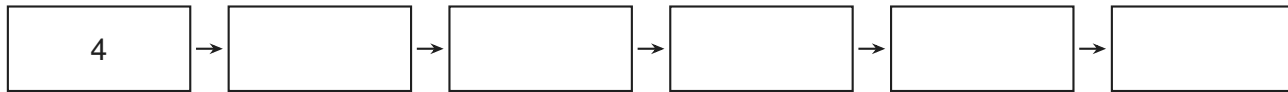
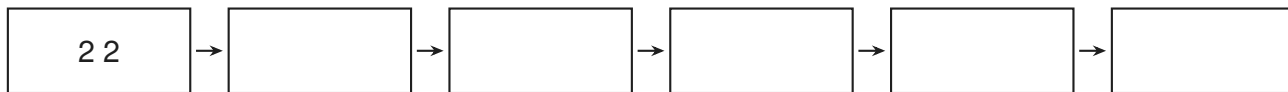
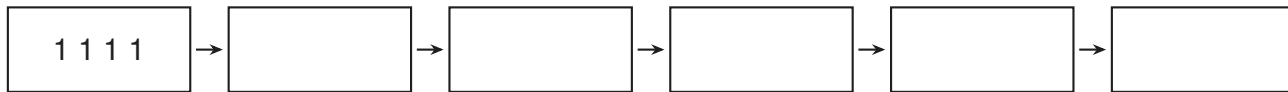


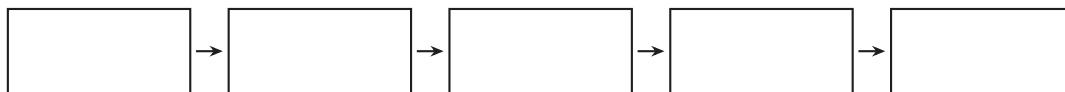
Problem 1: Take 4 counters and put them all in one pile. The move: take one counter from every pile, and put all the counters you took together as one new pile. A pile with no counters left is gone. After each move, put the piles in order from biggest to smallest and write the pile sizes in the next box, like 3 1. Keep going until you write an arrangement that is already in the strip. Circle the two matching boxes.



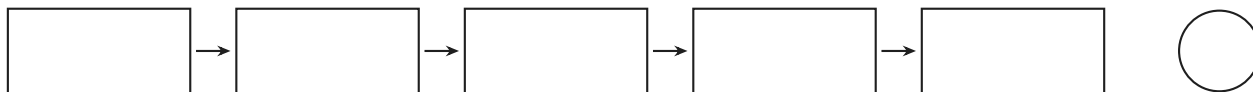
Problem 2: Do the same with 4 counters, starting from four piles of 1. Then do it again, starting from two piles of 2.



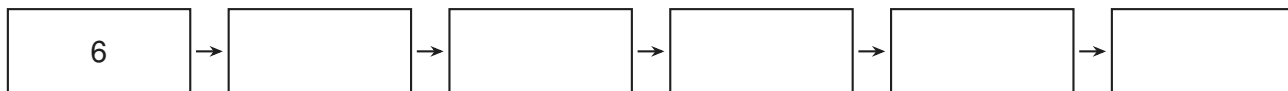
Problem 3: There are two more ways to put 4 counters in piles. Write each one in the first box of a strip, then run the move from it until an arrangement repeats.



Problem 4: Look at all five strips. Some arrangements come back again and again. Write them in these boxes in the order they come, one arrangement per box, and draw an arrow from the last one you wrote back to the first. In the circle, write how many arrangements are in the loop.



Problem 5: Take 6 counters, all in one pile. Run the move. This time keep going until a move gives back exactly the arrangement you just wrote, and then stop. An arrangement that the move does not change is called a stop. Circle the stop.



Problem 6: Find all the different ways to put 6 counters in piles. Write each way in its own box, biggest pile first. One way is 6, another is 5 1. Two ways with the same sizes in a different order count as the same way. When you cannot find any more, write how many ways you found in the circle.



Problem 7: Sort your ways by the size of the biggest pile. Write every way with biggest pile 6 in the first row of the table, every way with biggest pile 5 in the second row, and so on. If a row looks short, look for a way you missed. Write the new total in the circle.

biggest pile 6			
biggest pile 5			
biggest pile 4			
biggest pile 3			
biggest pile 2			
biggest pile 1			

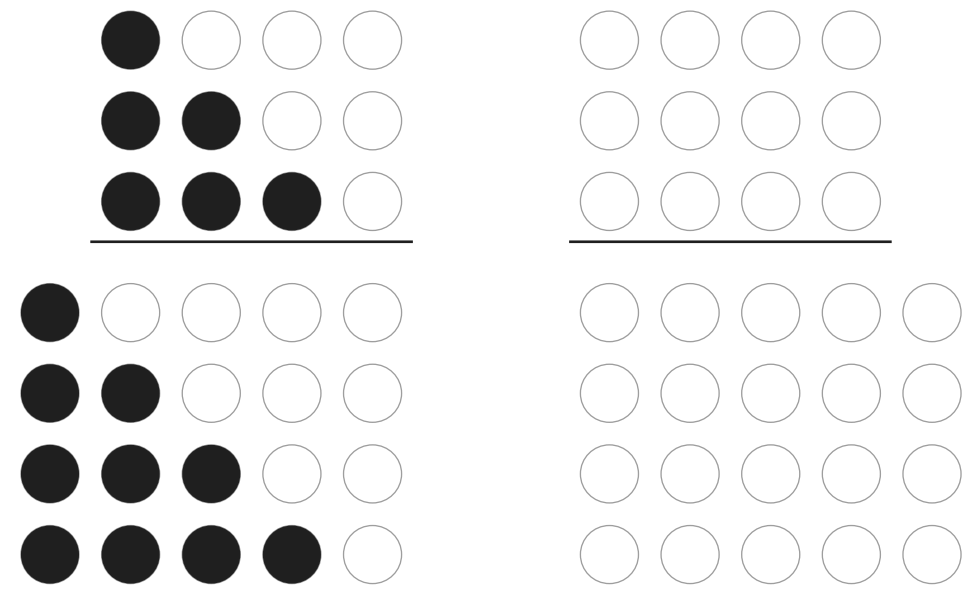


Problem 8: Run the move from every way in your table, one strip for each way, and share the strips among your table. Stop each run at the stop, and circle the stop the first time it appears. Count the arrows before the circled stop and write that number of moves in the small circle at the end of the strip.

[illegible]

Problem 9: Compare all the strips from Problem 8. Did every start reach the same stop? Which start took the most moves? Which start took the fewest moves, not counting the stop itself? Which arrangements appear in more than one strip? Write your answers on the lines.

Problem 10: Here is 3 2 1 drawn as towers of dots, one tower for each pile, tallest first. The move takes the top dot off every tower, and those dots become the new pile. Cross out the top dot of each tower, and draw the new pile as a tower of dots in the empty column. Then, in the second picture, draw the towers that are left and the new tower, tallest first. Explain to your table why the move can never change 3 2 1. Then do the same for 4 3 2 1 in the third and fourth pictures, and check with 10 counters whether 4 3 2 1 is a stop.



Problem 11: Build stairs of 1 and 2; then 1, 2 and 3; then 1, 2, 3 and 4; then up to 5; then up to 6. Each time, count the counters and write the number in the table, do the move, and write yes or no: is it a stop?

piles	counters	a stop?
1, 2		
1, 2, 3		
1, 2, 3, 4		
1, 2, 3, 4, 5		
1, 2, 3, 4, 5, 6		

Problem 12: Find all the ways to put 5 counters in piles. Write each way in a left box, biggest pile first. Then do one move from it with the counters and write the result in the right box. Circle any way that is a stop.

→	→
→	→
→	→
→	→

Problem 13: Start from 5 and run the move until an arrangement repeats. Then start from 2 1 1 1. Write the loop for 5 counters in the last row, with an arrow back to the first box, and write how many arrangements are in the loop in the circle.

5 → → → → → →
2 1 1 1 → → → → →
→ → → → → →

○

Problem 14: Now go backwards. Find a way to put 6 counters in piles so that one move gives 3 3. Find two different ways that give 4 2 after one move. Then try to find a way that gives 2 2 1 1 after one move. Look at your strips from Problem 8: which arrangements of 6 counters never appear after an arrow? Write them on the last line.

gives 3 3 after one move:

gives 4 2 after one move:

gives 2 2 1 1 after one move:

never after an arrow:

Problem 15: Take 7 counters, all in one pile, and run the move until an arrangement repeats. Then do the same with 8 counters starting from 8, and again with 8 counters starting from 5 3. Circle the loop in each strip and write how many arrangements are in it in the small circle.

